

HARNEET KAUR DEHIYA

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EDUCATION

Illinois Institute of Technology, Chicago, IL

August 2023 – May 2025

Master of Computer Science | GPA: 3.9

Relevant Coursework: Algorithms, Data Mining, Cloud Computing, Machine Learning, Mobile Application Development, Natural Language Processing, Data Preparation and Analysis, Healthcare AI

University of Mumbai, India

August 2019 – May 2023

Bachelor of Electronics and Telecommunication Engineering | GPA: 3.82

Relevant Coursework: Database Management Systems, Software Process Automation, Project Management

PROJECTS

Music Recommendation System | *Python, Librosa, Flask, Heroku, HTML/CSS*

[Link](#)

- Built and deployed an ML-powered web app for personalized song recommendations using real-time audio similarity.
- Extracted acoustic features (like MFCCs, Spectral Centroid, Zero Crossing Rate) from 1,200+ songs using Librosa.
- Applied K-Means clustering to segment songs into 26 groups; recommended 10 most similar tracks per query.
- Built a Flask-based backend and deployed the app on Heroku with a clean, responsive UI using HTML/CSS.

Stock Trading Agent with Reinforcement Learning | *Python, TensorFlow, Keras*

[Link](#)

- Developed a Double Deep Q-Learning (DDQL) model with LSTMs to simulate stock trading strategies.
- Created a custom trading environment with actions (Buy, Sell, Hold) and rules based on candlestick patterns.
- Designed a reward system that penalized risky/invalid actions and encouraged profitable decisions.
- Collected and preprocessed 10 years of stock market data (2004–2014) for training and evaluation.

Movie Review Sentiment Analysis (NLP) | *Python, scikit-learn*

[Link](#)

- Built a real-time sentiment classifier on 50K+ IMDB reviews using Naïve Bayes & Logistic Regression.
- Logistic Regression achieved 89% accuracy, Naïve Bayes: 85%; validated using precision, recall, F1-score, specificity.
- Implemented an interactive CLI tool with real-time predictions and confidence scoring.

OMOPSync – Healthcare Data ETL and NLP | *Python, R, PostgreSQL*

[Link](#)

- Built a Model Context Protocol server with ETL pipelines to support natural language-to-SQL queries on EHR data.
- Designed R pipelines to transform and load 10K+ patient records into PostgreSQL in OMOP common data model.
- Streamlined health data processing for research and ensured compatibility with standardized data models.

EXPERIENCE

Graduate Teaching Assistant | *CS422 – Data Mining*

College of Computing, Illinois Institute of Technology, Chicago, IL

Apr 2024 – May 2025

- Mentored 60+ students each semester in algorithms, data mining and machine learning techniques.
- Developed and deployed an automated grading pipeline in R for course labs, reducing evaluation time by ~45%.

Data Visualization Intern

GlobalShala – Remote

Sep 2022 – Oct 2022

- Analyzed marketing data and built Tableau dashboards to identify underperforming campaigns, cutting costs by 11%.
- Led a small project team by defining outlines, drawing insights, and ensuring on-time delivery of final reports.

SKILLS

- **Programming:** Python, R, SQL, Java, Dart, HTML/CSS, JavaScript
- **Frameworks & Tools:** scikit-learn, TensorFlow, Keras, PyTorch, LangChain, OpenAI API, Git, PostgreSQL, Tableau
- **Core Expertise:** Data Analysis, Machine Learning, Natural Language Processing, Deep Learning, Generative AI, LLMs